

● EASY

● ALGEBRA

● ANSWER KEY

Linear functions

05

10 answers · Rationale-rich · Teach from doc

Q1**A**

A. With each monthly deposit, the amount in Hana's bank account increased by \$ 25.

Choice A is correct. It's given that t represents the number of monthly deposits. In the given function $ft = 100 + 25t$, the coefficient of t is 25. This means that for every increase in the value of t by 1, the value of ft increases by 25. It follows that with each monthly deposit, the amount in Hana's bank account increased by \$ 25.

Choice B is incorrect. Before Hana made any monthly deposits, the amount in her bank account was \$ 100.

Choice C is incorrect. After 1 monthly deposit, the amount in Hana's bank account was \$ 125.

Choice D is incorrect and may result from conceptual errors.

Q2**1**

The correct answer is 1. It's given that the function f is defined by $fx = x + \frac{8}{11}$. Substituting $\frac{3}{11}$ for x in the given function yields $f\frac{3}{11} = \frac{3}{11} + \frac{8}{11}$, which gives $f\frac{3}{11} = \frac{11}{11}$, or $f\frac{3}{11} = 1$. Therefore, when $x = \frac{3}{11}$, the value of fx is 1.

Q3**69**

The correct answer is 69. The value of $f(20)$ can be found by evaluating the function $f(x) = \frac{7}{10}x + 55$ for $x = 20$. Substituting 20 for x in this function yields $f(20) = \frac{7}{10}(20) + 55$, or $f(20) = 69$. Therefore, the value of $f(20)$ is 69.

Q4

B

B. 250

Choice B is correct. Substituting 50 for x in the given function yields $h50 = 50 + 200$, or $h50 = 250$. Therefore, the value of $h50$ is 250.

Choice A is incorrect. This is the value of $h0$.

Choice C is incorrect. This is the value of $h9,800$.

Choice D is incorrect. This is the value of $h50,000$.

Q5

A

A. $h = 8s + 15$

Choice A is correct. It's given that the front of the roller-coaster car starts rising when it's 15 feet above the ground. This initial height of 15 feet can be represented by a constant term, 15, in an equation. Each second, the front of the roller-coaster car rises 8 feet, which can be represented by $8s$. Thus, the equation $h = 8s + 15$ gives the height, in feet, of the front of the roller-coaster car s seconds after it starts up the hill.

Choices B and C are incorrect and may result from conceptual errors in creating a linear equation. Choice D is incorrect and may result from switching the rate at which the roller-coaster car rises with its initial height.

Q6**C**

C. The snow leopard population in this area is predicted to be 550 in the year 2020.

Choice C is correct. It's given that $P(t)$ represents the population of snow leopards t years after 1990. $P(30) = 550$ corresponds to $t = 30$ and $P(t) = 550$. It follows that $t = 30$ corresponds to 30 years after 1990, or 2020. Thus, the best interpretation of $P(30) = 550$ is that the snow leopard population in this area is predicted to be 550 in the year 2020.

Choices A and B are incorrect and may result from reversing the interpretations of t and $P(t)$. Choice D is incorrect and may result from determining that 30 years after 1990 is 2030, not 2020.

Q7**C**

C. \$ 15,400

Choice C is correct. It's given that f_x is the total cost, in dollars, to lease a car from this dealership with a monthly payment of x dollars. Therefore, the total cost, in dollars, to lease the car when the monthly payment is \$ 400 is represented by the value of f_x when $x = 400$. Substituting 400 for x in the equation $f_x = 36x + 1,000$ yields $f_{400} = 36(400) + 1,000$, or $f_{400} = 15,400$. Thus, when the monthly payment is \$ 400, the total cost to lease a car is \$ 15,400.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q8**C****C.** 7

Choice C is correct. For the graph shown, the x -axis represents temperature, in kelvins, and the y -axis represents volume, in liters. Therefore, the estimated volume, in liters, of the hydrogen when its temperature is 500 kelvins is represented by the y -coordinate of the point on the graph that has an x -coordinate of 500. The point on the graph with an x -coordinate of 500 has a y -coordinate of 7. Therefore, the estimated volume, in liters, of the hydrogen when its temperature is 500 kelvins is 7.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Q9**D****D.** $f(x) = 39x$

Choice D is correct. An equation defining a linear function can be written in the form $fx = mx + b$, where m is the slope and b is the y -intercept of the graph of $y = fx$ in the xy -plane. It's given that the graph of $y = fx$ has a slope of 39, so $m = 39$. It's also given that the graph of $y = fx$ passes through the point $0, 0$, so $b = 0$. Substituting 39 for m and 0 for b in $fx = mx + b$ yields $fx = 39x + 0$, or $fx = 39x$. Thus, the equation that defines f is $fx = 39x$.

Choice A is incorrect. This equation defines a function whose graph has a slope of -39, not 39.

Choice B is incorrect. This equation defines a function whose graph has a slope of $\frac{1}{39}$, not 39.

Choice C is incorrect. This equation defines a function whose graph has a slope of 1, not 39, and passes through the point $0, -39$, not $0, 0$.

Q10

B

B. $f(x) = 1.8x$

Choice B is correct. It's given that the table represents the relationship between the area x , in square feet, of the roof of a shopping center and the amount of water $f(x)$, in gallons, drained from the roof into the gutters. Every choice represents this relationship with an equation defining f in the form $f(x) = mx$, where m is a constant rate of change. The value of m can be determined by dividing both sides of the equation by x . Each of three pairs of x and $f(x)$ in the table yield $m = 1.8$, since $\frac{4,536}{2,520} = 1.8$, $\frac{6,804}{3,780} = 1.8$, and $\frac{9,072}{5,040} = 1.8$. Therefore, the equation $f(x) = 1.8x$ could define f .

Choice A is incorrect. For the roof with an area of 2,520 square feet, this equation would yield $0.6(2,520)$, or 1,512, gallons, not the 4,536 gallons shown in the table.

Choice C is incorrect. For the roof with an area of 2,520 square feet, this equation would yield $2,268(2,520)$, or 5,715,360, gallons, not the 4,536 gallons shown in the table.

Choice D is incorrect. For the roof with an area of 2,520 square feet, this equation would yield $4,536(2,520)$, or 11,430,720, gallons, not the 4,536 gallons shown in the table.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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